

ADVERSARIAL VULNERABILITY AUDIT REPORT

Model

demo_model

2026-09-26 22:47 UTC

8.9

CRITICAL

KAAL VULNERABILITY SCORE (0 = Robust / 10 = Catastrophic)

AUDIT PARAMETERS

Attack	Epsilon	Steps	Success Rate
FGSM	0.03	—	—
PGD	0.03	20	—

OVERALL FINDINGS

8.9 Critical

Fgsm Susceptibility	10.0
Pgd Susceptibility	10.0
Empirical Robustness	10.0
Adversarial Overconfidence	0.8

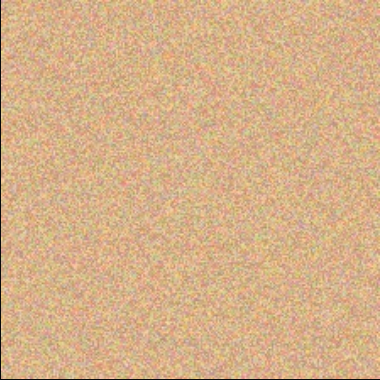
RECOMMENDED ACTIONS

1. Apply input preprocessing: JPEG compression or random noise addition before inference.
2. Implement adversarial training: retrain model with PGD-generated examples included in training set.

ATTACK PARAMETERS

Epsilon (ϵ)	0.03
Success Rate	—
Avg Confidence Drop	+0.0331

ADVERSARIAL EXAMPLE



Predicted: class 640 (6.73%)

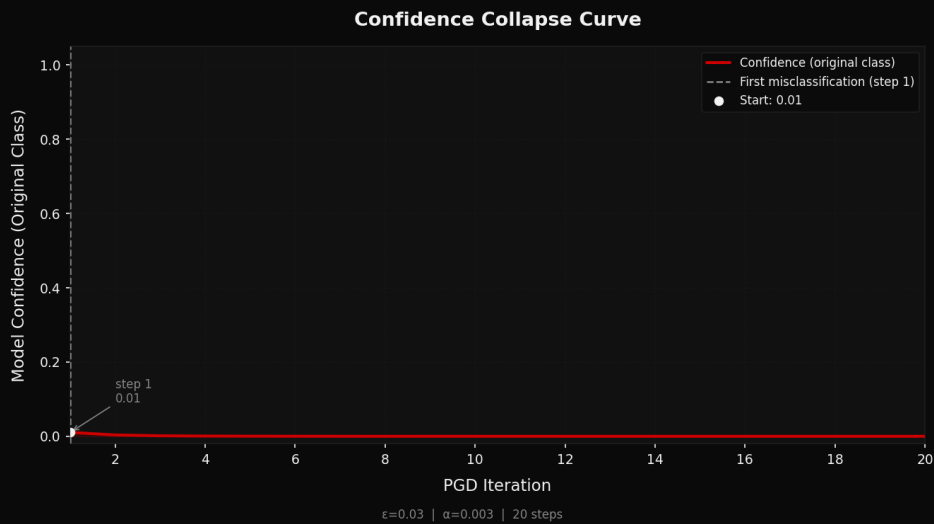
FINDING

Model prediction changed from class 892 to class 640 under perturbation magnitude $\epsilon=0.03$; original confidence 0.03 reduced to 0.07.

ATTACK PARAMETERS

Epsilon (ϵ)	0.03
Alpha (α)	0.003
Steps	20
Success Rate	—
Steps to Success	1

CONFIDENCE COLLAPSE CURVE



FINDING

PGD at perturbation magnitude $\epsilon=0.03$, $\alpha=0.003$ caused misclassification from class 892 to class 78 at step 1 of 20; original confidence 0.03 reduced to 0.02.

Patch attack not run.

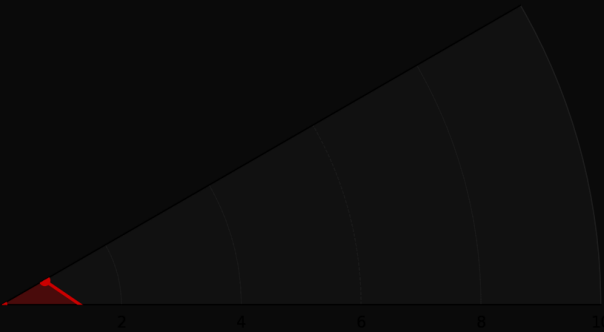
Physical robustness test not run.

FGSM
10.0

demo_model KVS 8.9 [Critical]

PGD
10.0

Overconf
0.8



Physical
0.0

Empirical
10.0

Black-box
0.0

10.0	Fgsm Susceptibility
10.0	Pgd Susceptibility
10.0	Empirical Robustness
0.8	Adversarial Overconfidence

ENVIRONMENT

KAAL Version	1.0.0
Python	3.10.5
Platform	Windows 10
Audit Duration	46s
PyTorch	2.2.0+cpu

MODEL INFORMATION

Model Path	demo_model.pt
Model Name	demo_model
Framework	pytorch
Input Shape	[3, 224, 224]
Num Classes	1000

DATASET STATISTICS

Dataset Path	demo_images
Total Images	10
Formats	.jpg=10

FULL ATTACK PARAMETERS

FGSM	$\epsilon=0.03$	steps=	success=
PGD	$\epsilon=0.03$	steps=20	success=